

ICT703 Programming

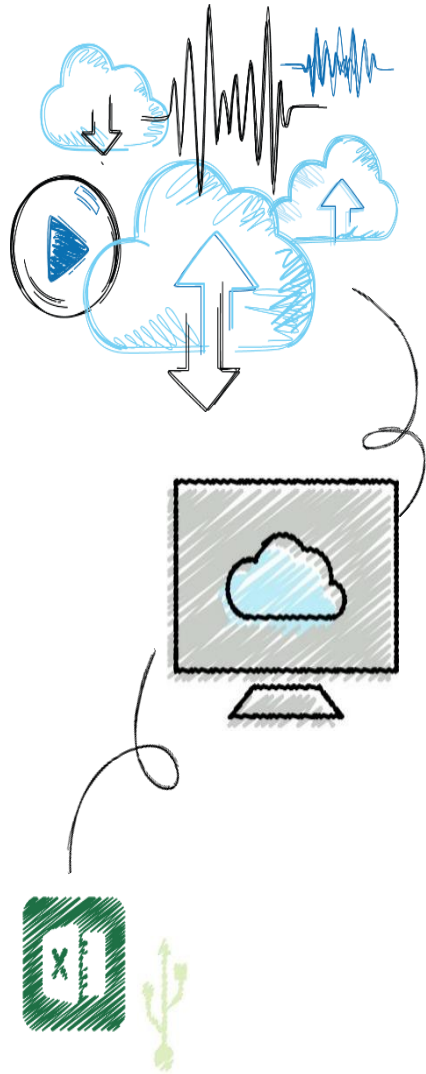
Module 6

Tuples



Objectives

1. Describe a Tuple Data Type and its uses
2. Passing multiple values at once





Programming Concepts

Four control flow

- Sequence
- Iteration
- Selection
- Abstraction – methods/classes

Data



Introduction

- A **tuple** resembles a list, but is immutable
 - Sequential collections (strings, lists, etc) can be immutable or mutable
 - Mutable – the elements can be changed
 - Lists are mutable. We can have:
 - `listExample = ['Andrew', 'Peter', 'Sam']`
 - `listExample[0] = 'Fred'` `#this changes the list element Andrew to Fred`
 - Immutable – the elements can NOT be changed
 - Strings are Immutable. We will get an error if we try:
 - `stringExample = "Andrew"`
 - `stringExample[0] = 'b'`
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Tuples - Examples

- A **tuple** resembles a list, but is immutable
 - Indicate by enclosing its elements in ()

```
fruits = ("apple", "banana")  
print(fruits)  
(‘apple’, ‘banana’)
```

```
meats = ("fish", "poultry")  
print(meats)  
(‘fish’, ‘poultry’)
```

```
food = meats + fruits  
print(food)  
(‘fish’, ‘poultry’, ‘apple’, ‘banana’)
```



Tuples are good for records

- A tuple can be used to pass records to and from a (relational) database
 - `julia = ("Julia", "Roberts", 1967, "Duplicity", 2009, "Actress", "Atlanta, Georgia")`
 - Unlike a dictionary we do not carry around the names of the fields but reference them like a list
 - `print(julia[1])`
 - Roberts
 - A tuple is good to store multi- values when you need to get to the component parts. For example a date:
 - `today = (24, 9, 2012)`
 - `(day, month, year) = today`
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Tuples are good for returning multi values

- Most functions return 0 or 1 value but many times we might want to create a function that returns a few values. Tuples are really good for this
- For example, imagine there was a function that took a name and returned the person's age, height and weight
- `(age, height, weight) = getDetails('fred')`



Summary

- A tuple is an immutable object - you can't change the values
- It behaves like a list – you use [0] to access elements
- Very useful for records of information and returning information from functions